

Grid Coordination

For Residential Customers

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Two Grid Coordination Problems

"Macro" Grid Coordination

Managing the Grid

- Highly dynamic prices
- Demand-response events
- Functional control of devices
- Grid events (FlexAlerts)

"Micro" Grid Coordination

Managing the Distribution Network

- Functional control at the device level (EVSE, DER inverter)
- Dynamic control of customer-site service level (import & export)
- Can't do much of this today: static sizing based on predicted worst-case load

Trends: Grid

- Increasing agreement that **highly dynamic pricing** is superior to traditional DR events
- CPUC & CEC vision: transition to highly dynamic pricing, with LSE/IOU pilots underway
- The number of flexible-demand appliances per household is **growing**
 - Multiple appliances need to be managed "in concert"
 - Unlikely that all appliances share the same manufacturer
- Grid coordination is transitioning from the **device** to the **site**
 - Set a dynamic power import/export limit on the site, not per device
 - The site manages loads & appliances via PCS & HEMS (NEC-705.13 & UL-3141)
- Site power & energy management will most likely occur **locally, not in the cloud**

Trends: Smarter Infrastructure

Smart Meters

- Improved sensors with high sample-rate ADCs
- Network connectivity to the home LAN and Internet
- Third-party meter apps and platform standards (GEISA)

Smart Electric Panels

- Revenue-grade metering on I/O lugs & every circuit
- Relays on each circuit
- Network connectivity: Ethernet and Wi-Fi
- Integrated Power Control Systems (NEC-705.13 & UL-3141)
- Integrated Home Energy Management System (HEMS)
- Gateway processor running Linux

Trends: Matter

- Widespread agreement & adoption of **Matter** as the open standard for smart-appliance & IoT interoperability
 - Apple, Google, Amazon, LG, 800+ others
- However, Matter **does not** provide grid-to-customer coordination
- Matter alone cannot provide a complete grid-to-customer flexible demand solution

The Combination

- **HEMS + OpenADR 3.1** can shift appliance energy usage via Matter
- **PCS** can attenuate appliance power via Matter
 - Better customer experience than power-cycling via relays
 - Relays provide a fail-safe backstop

Grid Coordination with Matter

Use **OpenADR 3.1** to obtain DR events and electricity prices from the cloud/Internet to the home

Use **Matter** within the home to control the power and shift energy of appliances



Two Protocol Layers

Layer	Protocol	Scope
Grid-to-home	OpenADR 3.1	Prices, DR events, power limits
Within-home	Matter	Device control, energy management

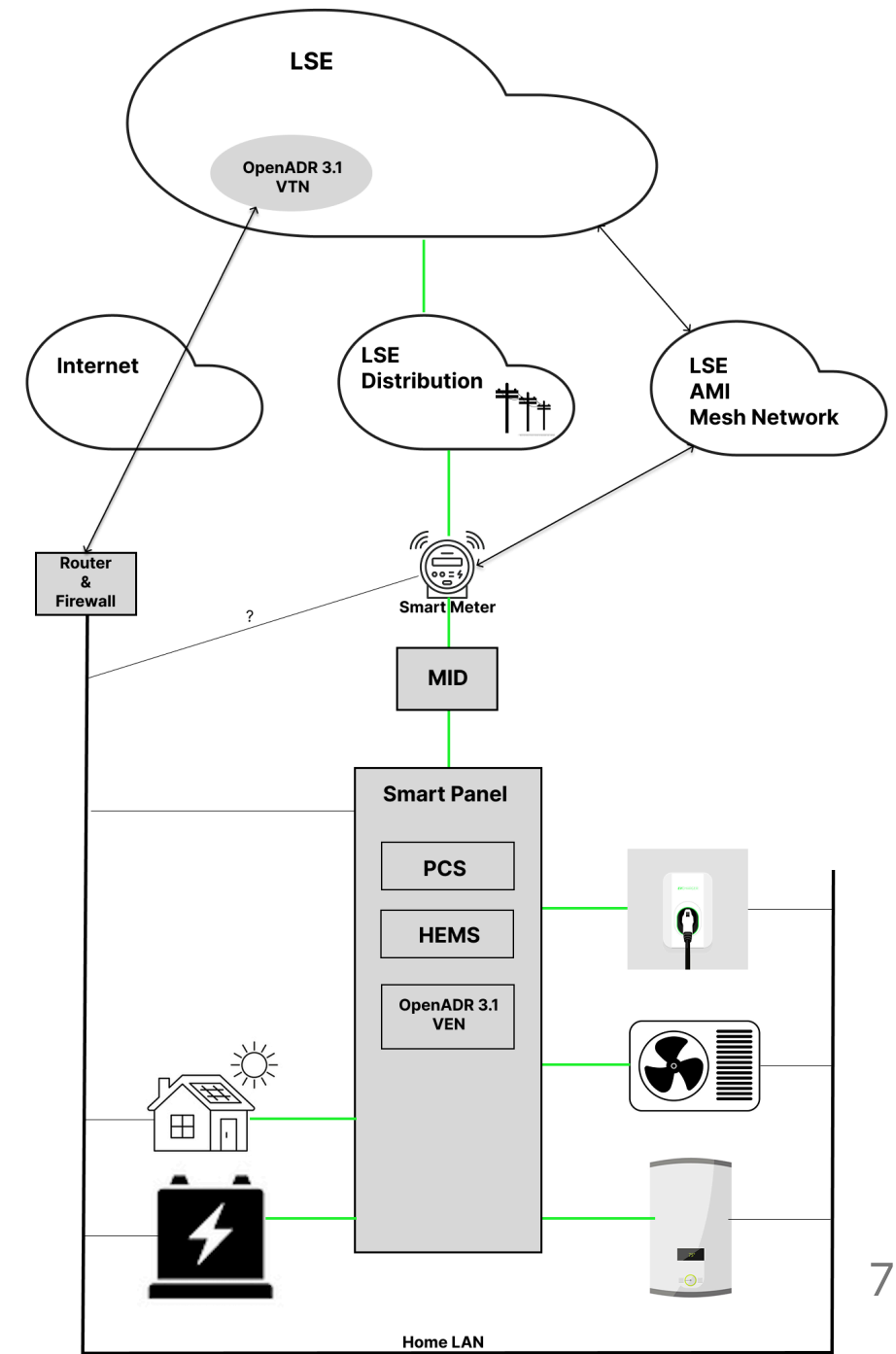
The Future Residential Utility Customer

Smart Meter capabilities:

- Power import & export measurement
- Grid health monitoring via high sample-rate measurements
- Communication with LSE via private wireless mesh
- Connection to home LAN and Internet
- Communication with home energy infrastructure: smart panels, DERs, EVSE, HVAC, water heaters

Smart Panel capabilities:

- Integrated MID (Microgrid Interconnect Device)
- Revenue-grade metering on every circuit with relays
- Integrated PCS (NEC-705.13 & UL-3141)



Smart Meter and Smart Panel Integration

Use Cases (prioritized)

- 1. Meter sets Dynamic Service Rating (DSR) on Panel PCS
- 2. Meter subscribes to circuit power measurements from panel
- 3. DR events from meter to panel
- 4. Panel subscribes to high sample-rate waveform analysis from meter

Protocol Options

Protocol	Strengths	Timeline
Matter	Standard, local, appliance integration	Key clusters in 1.5-1.6
eBus	Simple, proven (MQTT, Homie v5, mDNS)	Working today

Matter Taxonomy & Roles

Cluster Server

- The device you want to control or read state from

Controller (Client)

- Entity that talks to the cluster server
- Can be simple bindings (e.g., light switch controlling a bulb)

Commissioner

- Controls pairing and owns the fabric
- Assigns Node Operational Certificates (NOCs)
- Devices with NOCs signed by the same Root CA trust each other
- Sets Access Control Lists (ACLs): which node can do what
- ACLs can be granular: specific node, specific attribute, specific endpoint

Functional Control and Demand Response

- Device behavior is ultimately altered via **functional-control commands**
- Optimally translating a DR event into a functional-control command requires **detailed knowledge & context** of the customer
- The further from the customer this translation occurs, the **less context** is available
- Local optimization **cannot** occur at the grid/LSE or aggregator level

Key Principles

- The "location" where translation occurs must be **flexible** — customer's choice
- The DR event **must** be accessible at: the device, a local HEMS, or a delegated ASP
- This translation **must not** be locked within the manufacturer's cloud

Local HEMS is the best location to translate price & DR events to functional-control commands